Department of Electronic & Telecommunication Engineering
University of Moratuwa



EN4431 - Analog IC Design

Assignment 04
Integrated Layout Design of an Integer-N Phase Locked Loop

Nethmal W D I - 210421E

1 Introduction

In this stage, the previously designed PLL blocks were treated as hard IPs and integrated into a single top-level layout. The main focus was to choose a suitable floorplan, reduce the overall area, complete routing between blocks, and verify the final layout using DRC and LVS.

The main blocks integrated in the top-level PLL layout were the Phase Frequency Detector (PFD), Charge Pump (CP), Loop Filter, Voltage-Controlled Oscillator (VCO), and the Divider. The final layout was designed in the SKY130 process using Cadence Virtuoso.

2 Floorplan and Placement Justification

The floorplan was selected by considering the signal flow, sensitive analog nodes, routing length, and area. Since this is a mixed-signal PLL, the VCO and loop filter require more careful placement than the more digital blocks such as the divider.

The main placement decisions were as follows:

- The PFD was placed close to the Charge Pump to reduce the routing length of the UP and DOWN control signals.
- The Charge Pump was placed near the Loop Filter because the charge pump output is a sensitive analog current-output node.
- The Loop Filter was placed close to the VCO control input to reduce parasitic capacitance and noise pickup at the VCO control node.
- The VCO was placed away from unnecessary long digital routing to reduce substrate and supply noise coupling.
- Power and ground routing were arranged to provide stable supply connections to the analog and digital sections.

3 Loop Filter Design Choice

The loop filter was one of the most important layout decisions in the final integrated PLL layout. Ideally, the loop filter resistors should be implemented using poly resistors because they provide much smaller area for the required resistance values. However, during the final layout implementation, a DRC error occurred with the poly resistor implementation. Due to the limited time available before submission, this DRC issue could not be fully resolved. And it seems to be common, and we will share the final LVS cleared layout as soon as possible.

Therefore, the final submitted layout used metal1 resistors for the loop filter. This allowed the layout to be completed and verified more quickly, but it increased the total layout area significantly.

The loop-filter capacitors were planned to be implemented externally because the required capacitance values are relatively large for full on-chip integration in SKY130. Implementing these capacitors on chip would significantly increase the layout area and make the PLL less suitable for compact Tiny Tapeout integration. By using external capacitors, the on-chip area can be reduced

while also allowing more flexibility to tune the loop-filter response during testing. This is useful because the PLL bandwidth, damping factor, and lock behaviour can be adjusted more easily by changing the external capacitor values without redesigning the chip layout.

3.1 Metal1 Resistor Implementation

The final layout used metal1 resistors in the loop filter and it was sized around 80 um*165 um. The advantage of this choice was that it avoided the unresolved DRC issue associated with the poly resistor implementation. It also allowed the final top-level layout to be completed within the available time.

However, the disadvantage is that metal resistors require a much larger physical area to obtain the required resistance. Since metal has a lower sheet resistance than high-resistance poly, a much longer metal path is required to obtain the same resistance value. As a result, the loop filter occupied a large portion of the final PLL layout, as shown below in Fig. 1.

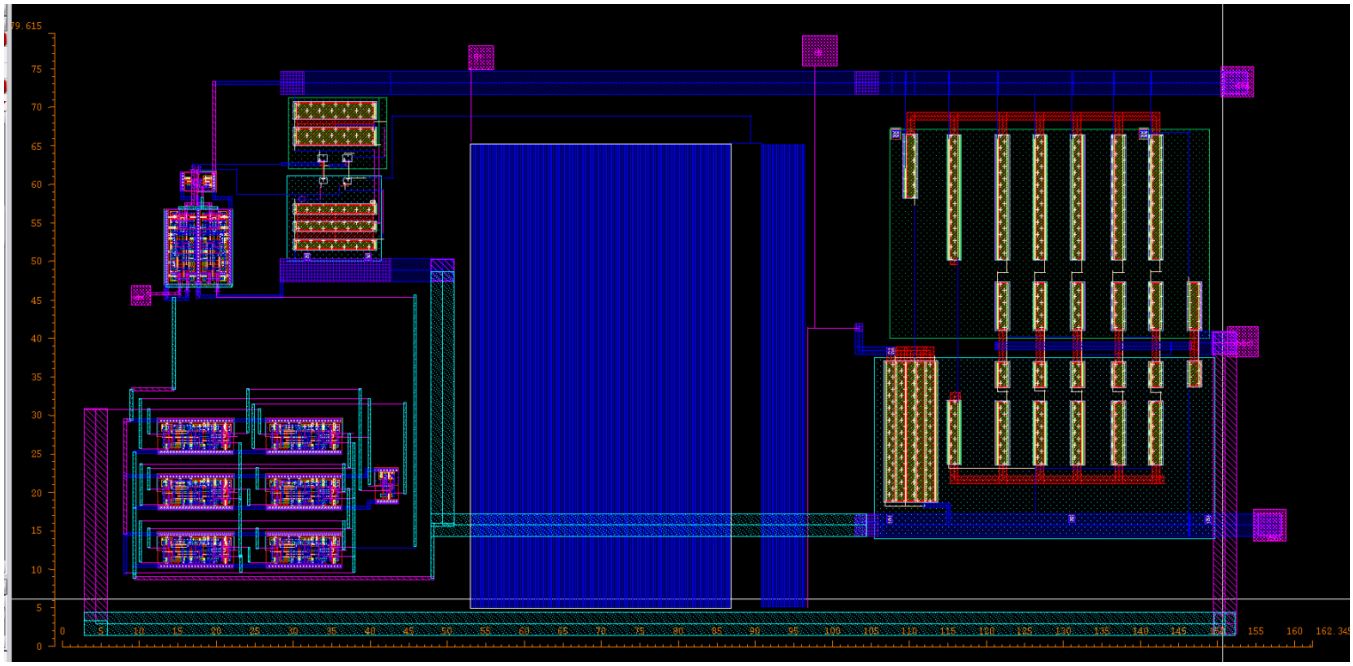


Figure 1: Final integration with metal resistors

3.2 Poly Resistor Alternative

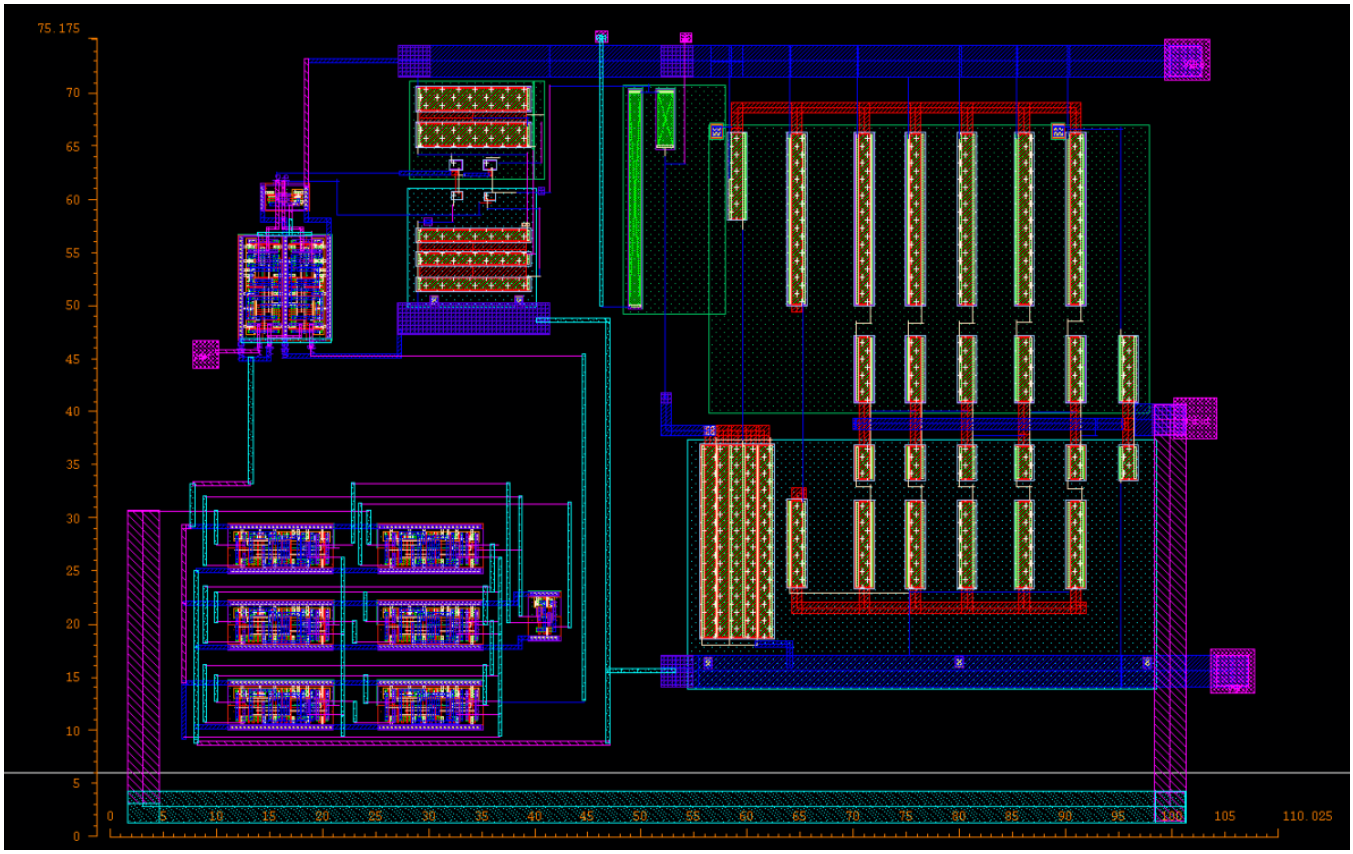


Figure 2: Final integration with poly resistors

The poly resistor implementation would have been more area-efficient. This would allow the same loop filter resistance values to be implemented using a shorter and more compact structure. The estimated layout size using the poly resistor implementation was approximately:

$\mathrm{A}_{\mathrm{poly}} = 75 \times 110 = 8250\ \mu\mathrm{m}^2$

3.3 Area Comparison

The current final layout using metal1 resistors has an approximate dimension of:

$$A_{\mathrm{metal}} = 165 \times 80 = 13200\ \mu\mathrm{m}^2 \tag{1}$$

Then the reduction of area is approximately,

$$\Delta A = 13200 - 8250 = 7800\ \mu\mathrm{m}^2, \quad \frac{13200}{8250} \approx 1.6 \tag{2}$$

The percentage area reduction possible by using the poly resistor version is 39% Therefore, replacing the metal1 resistors with properly DRC-clean poly resistors in the final tapeout version can reduce the layout area largely.

```
Total CPU Time                  : 0(s)
Total Real Time                 : 0(s)
Peak Memory Used                : 18(M)
Total Original Geometry         : 2941(7814)
Total DRC RuleChecks            : 578
Total DRC Results               : 0 (0)
Summary can be found in file PLL.sum
ASCII report database is /home/user19/PLL_421E/drc/PLL.drc_errors.ascii
Checking in all SoftShare licenses.
```

Figure 3: DRC results of the layout

4 DRC and LVS Verification Observations

The final integrated layout was verified using the SKY130 physical verification flow. During the loop-filter implementation, three resistor options were tested: high-resistivity poly resistors, normal poly/diffusion-based resistors, and metal resistors. The observed verification results are summarized below:

- **High-resistivity poly resistors**
  - DRC: Failed
  - LVS: Passed
- **Normal poly/diffusion-based resistors**
  - DRC: Passed
  - LVS: Failed
- **Metal resistors**
  - DRC: Passed
  - LVS: Failed

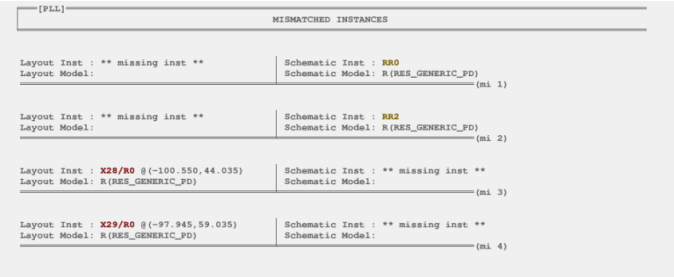


Figure 4: LVS error in POLY resistors

From these observations, the high-resistivity poly resistor implementation was favourable from an LVS point of view, but it produced DRC violations that could not be resolved within the available time. On the other hand, the normal poly/diffusion-based and metal resistor implementations passed DRC, but produced LVS mismatches. Therefore, the final resistor choice involved a trade-off between DRC-clean implementation and LVS compatibility.

A possible reason for the LVS failure is a mismatch between the schematic netlisting format and the layout extraction format used by the SKY130 LVS flow. This may be due to a local configuration issue in the CDF/auCdl setup, or a consistency issue between the resistor schematic model and the LVS extraction device template. This needs to be further investigated before the final tapeout submission.

One observed case is shown below. In the schematic auCdl netlist, the `res_generic_pd` resistor is netlisted with a substrate/body connection using the `$SUB` parameter:

```
RR0 net06 C1 res_generic_pd $SUB=VDD! r=3.94K w=1u l=20u
RR2 net06 C2 res_generic_pd $SUB=VDD! r=656.667 w=1.5u l=5u
```

This indicates that the schematic-side representation includes the resistor terminals together with a body/substrate reference. For a p-diffusion resistor, this body connection is physically meaningful because the resistor is formed with respect to the well/substrate region, and the body is normally biased to `VDD!`.

However, the extracted layout netlist represents the same resistor using a two-terminal LVS device template:

```
*.DEVTMPLT 0 R(RES_GENERIC_ND) diff_res np_term(P) np_term(N)
*.DEVTMPLT 1 R(RES_GENERIC_PD) diff_res pp_term(P) pp_term(N)

.subckt res_generic_pd$$63 1 2
** N=12 EP=2 FDC=1
R0 2 1 0 $[RES_GENERIC_PD] $X=0 $Y=-40 $dt=1
.ends res_generic_pd$$63
```

Here, the extracted layout resistor has only two electrical terminals, corresponding to the positive and negative resistor terminals. The body/substrate terminal is not included as a separate LVS terminal in the extracted layout device. Therefore, LVS may interpret the schematic-side and layout-side resistors as different device representations even though they correspond to the same physical resistor structure.

A similar issue was also observed for the metal-resistor implementation. In that case, the extracted metal resistor network appeared to create unexpected connectivity during LVS comparison. This suggests that the metal-resistor extraction and schematic representation need to be checked carefully before final submission.

Therefore, the LVS mismatch is most likely not due to an intentional schematic/layout design difference, but due to the difference between the schematic auCdl resistor representation and the layout extraction device template. For the final tapeout version, this issue should be resolved by using a resistor device whose schematic and layout views are LVS-compatible, or by correcting the local CDF/auCdl and LVS rule configuration so that the schematic and extracted layout netlists use the same resistor terminal definition and parameter format.

References

[1] R. Jacob Baker, CMOS: Circuit Design, Layout, and Simulation, 3rd ed., Wiley, 2010.

[2] B. Razavi, Design of Analog CMOS Integrated Circuits, McGraw-Hill Education.

[3] B. Razavi, RF Microelectronics, Pearson Education.